🎯 Mini Project 2 🎯

Kafka Integration on Telco Customer Churn

You will extend your Mini Project 1: Productionising Telco Churn Prediction by adding Kafka + batch / streaming pipelines.

Dataset reference: [Telco Customer Churn (Kaggle)]

Dive into Real World ML.

Producer side:

Imagine Telco company systems emitting customer events (sign-ups, payments, contract renewals, churn flags, etc.). These will be streamed or sent in batches / streams to Kafka.

Consumer side:

Your ML pipeline (built in Mini Project 1) should read these messages, run feature transformations + churn predictions, and publish results to another topic.

Kafka Topics

telco.raw.customers

Input stream of customer records/events.

telco.churn.predictions

Output churn probability or classification.

telco.deadletter

(Optional) For invalid records.

Message Schema

📌 Map directly from the dataset fields:

```
{
  "customerID": "7590-VHVEG",
  "gender": "Female",
  "SeniorCitizen": 0,
  "Partner": "Yes",
  "Dependents": "No",
  "tenure": 1,
  "PhoneService": "No",
  "MultipleLines": "No phone service",
  "InternetService": "DSL",
  "OnlineSecurity": "No",
  "OnlineBackup": "Yes",
  "DeviceProtection": "No",
  "TechSupport": "No",
  "StreamingTV": "No",
  "StreamingMovies": "No",
  "Contract": "Month-to-month",
  "PaperlessBilling": "Yes",
  "PaymentMethod": "Electronic check",
  "MonthlyCharges": 29.85,
  "TotalCharges": 29.85,
  "Churn": "No",
  "event_ts": "2025-10-03T04:00:00Z" // event timestamp
}
```

- **Key:** customerID
- **Value:** JSON object like above.
- For batch runs, events may come from Telco Churn CSV file chunks.
- For streaming, generate synthetic continuous events with random sampling from dataset rows.

Pipelines

🔹 Producer (producer.py)

Single script, configured via argparse with --mode {streaming,batch}.

Streaming mode:

- Continuously samples rows from dataset (with random event_ts) and pushes to telco.raw.customers.
- Configurable rate (--events-per-sec).

Batch mode:

- Reads the full CSV and sends messages in chunks (--batch-size).
- Supports checkpoint/resume to avoid duplicates on restart.

🔹 Consumer (consumer.py)

Single script, configured via argparse with --mode {streaming,batch}.

Streaming mode:

- Reads events from telco.raw.customers.
- Runs the Mini Project 1 trained model (scikit-learn / Spark MLlib).
- Publishes churn probability results into telco.churn.predictions.

Example output:

```
{
  "customerID": "7590-VHVEG",
  "churn_probability": 0.82,
  "prediction": "Yes",
  "event_ts": "2025-10-03T04:00:00Z",
  "processed_ts": "2025-10-03T04:00:01Z"
}
```

Batch mode:

- Consumes a defined offset/time window (e.g., 1,000 records).
- Runs batch predictions.
- Optionally outputs a summary (e.g., churn % per contract type, top-K customers by churn risk).

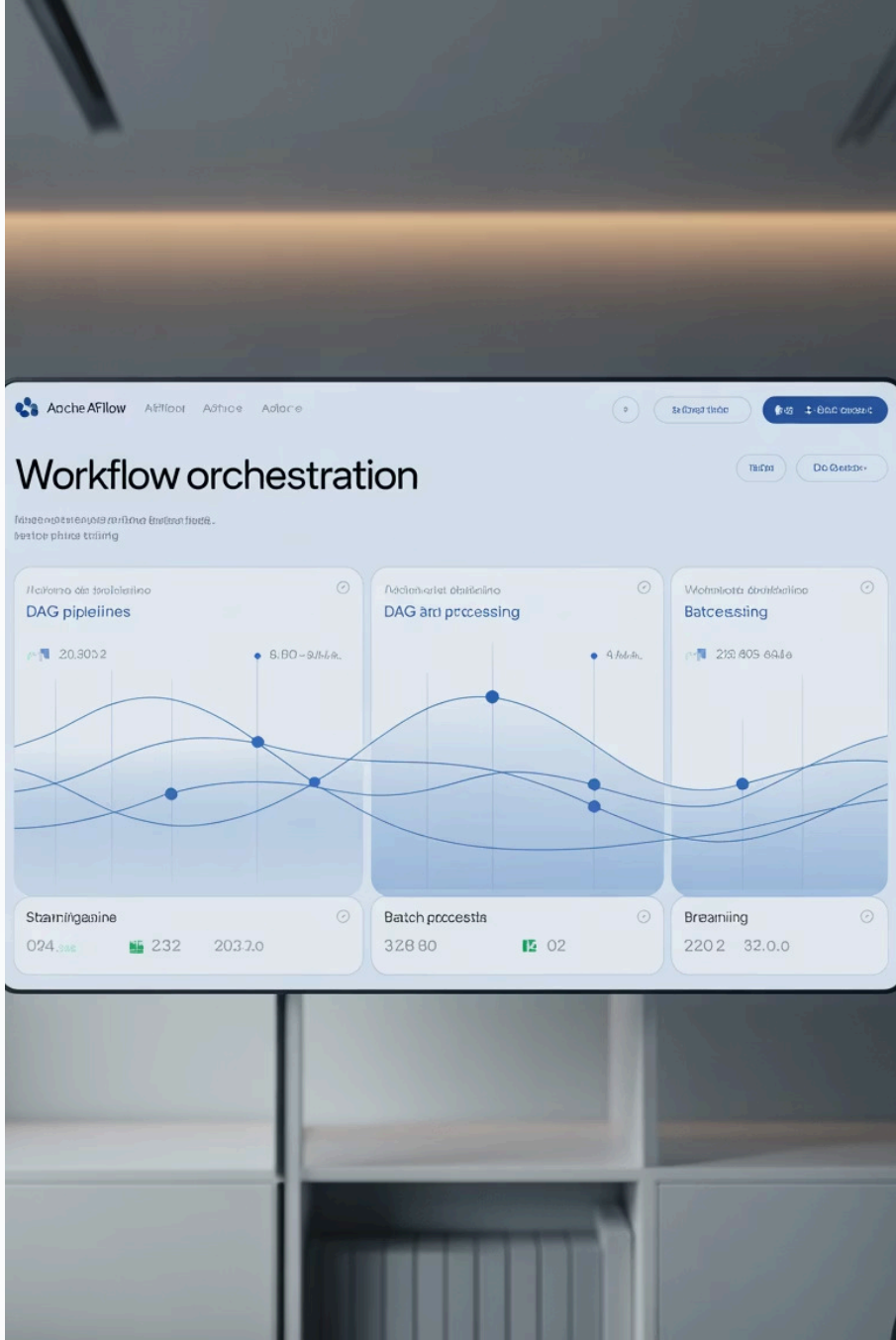
Bonus: Airflow DAGs (+10 Marks)

Streaming DAG

Health check + long-running consumer.

Batch DAG

Hourly/daily pipeline → producer → consumer → summary.



Deliverables

- **Kafka Integrated Pipeline (including 2 scripts):**
  - producer.py (supports streaming + batch)
  - consumer.py (supports streaming + batch)
- **Logs & screenshots: proof of messages flowing from dataset → Kafka → predictions.**
- **Configs for topic names, broker, batch sizes.**
- **(Bonus) Airflow DAGs & screenshots.**

Evaluation Rubric (100 + bonus)

20

Producers

✅ streaming + batch correctness.

40

Consumers

✅ streaming + batch with predictions.

20

Integration & reliability

✅

10

Testing & Observability

✅

10

Docs & repo hygiene

✅

+10

Bonus

⭐ Airflow orchestration.